The Constraint Layer: Autonomous Agents, Deterministic Ledgers, and the Missing Substrate of Machine Consequence

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Abstract

An autonomous software agent can now decide, but it cannot bind. A current model will plan a transaction, draft a contract, or instruct a payment, yet nothing it emits is by itself accountable: its behaviour is non-deterministic, it holds no stake, it cannot prove that it stayed within the authority it was given, and it cannot make any outcome irreversible. The deficit is structural rather than a matter of scale— more capable models decide better; they do not thereby become bindable. We argue that the missing faculty—rule-bound, provable, settleable consequence—is precisely what a particular kind of public ledger supplies, and that the two are complements rather than competitors: generative systems are open-ended and unconstrained exactly where ledgers are rigid and rule-governed, and each is of little use to consequential autonomy without the other. We give the anatomy of the gap; state the five properties a ledger must hold to act as a *constraint layer* for agents—provably secure consensus; deterministic settlement under which a locally checked action equals the globally settled one; proofs expressive enough to carry private compliance predicates; an economy that funds neutrality without an owner; and rules that can amend themselves—and show why the dominant ledger designs satisfy at most some of them. We then read one deployed architecture against the five as a case study, in design terms rather than market terms, and describe the form agent commerce takes once authority is compiled into a checkable policy and settlement doubles as audit. Two technologies usually filed together as a fad are in fact the two halves of one missing primitive: machines that decide, and a substrate on which their decisions can carry consequence. The decision is becoming abundant; the consequence is not; and building the second is the harder, and the more neglected, half.

Keywords. autonomous agents; verifiable computation; deterministic settlement; zero-knowledge proofs; smart contracts; delegation; AI safety.

1 Introduction

The hard problem of an autonomous agent is not that it might think poorly. It is that it can act without being bound. A language model wired to tools will read a market, choose a counterparty, size a position, and issue the instruction to move funds—and at no point in that sequence is there anything that holds it to what it was permitted to do, records what it actually did in a form a

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stranger can check, or makes the result final. Capability research is pouring effort into the first half of autonomy, the deciding. The second half, the part that lets a decision *count*, has no comparable program, because it was never a property of the model in the first place.

That second half has a name in economics it predates: the machinery by which parties who do not trust each other can nonetheless transact. Institutions exist to lower the cost of doing so [1], and trust is the lubricant that lets the whole system move [2]. For human actors the machinery is ambient—contracts, registries, courts, reputations, the threat of consequence—and an autonomous agent inherits none of it. It has no reputation that a bad act would damage, no assets a court could reach, no identity that persists with liability across runs. The literature on agentic systems has catalogued the resulting hazards [3, 4]; we read them as symptoms of a single absence. An agent is a faculty of decision without a faculty of consequence.

This article makes three moves. First (§2), it gives the anatomy of that absence: the specific, structural things an agent lacks the moment its decisions are supposed to bind other parties—and argues that none of them is closed by making the model larger. Second (§3–§4), it identifies the complement. A public ledger of the right kind is not another decider competing with the model; it is a substrate of consequence— deterministic where the agent is non-deterministic, rule-governed where the agent is open-ended, able to make an outcome provable and final where the agent can only assert. The fit is close enough to look designed: the agent supplies what the ledger cannot (open-ended decision), the ledger supplies what the agent cannot (bound, provable, settled consequence). But "a ledger" is not enough; only a ledger with five specific properties can hold an agent without reintroducing the trusted third party it was meant to remove, and the best-known designs hold at most some of the five. Third (§5–§6), it reads one deployed architecture against those five requirements as a worked case—on documented design, not market performance—and describes the shape commerce takes when an agent's authority is compiled into a predicate its counterparties can check and settlement is itself the audit trail.

The claim is deliberately narrow and falsifiable in form. It is not that blockchains are good, or that any token will appreciate; the case study (§5) is explicit that architecture is not adoption, and §7 states what would refute the thesis. The claim is that the pairing of generative decision with verifiable, deterministic consequence is a genuine technical direction rather than a slogan, that it is forced by the way autonomy is arriving, and that most of the engineering difficulty lives on the consequence side, which the present enthusiasm for ever-larger deciders is not addressing.

2 The Unbound Agent

What, precisely, does an autonomous agent lack the instant its outputs are meant to move money, bind a counterparty, or carry legal weight? Six things, none of them an intelligence deficit.

Non-determinism. A model's output is a sample, not a function. Re-run it and the answer can change; the same is not acceptable of a payment or a contract execution, where the property required is that the operation mean exactly one thing and that any observer recompute it identically. Determinism is not a quality a more capable sampler converges to; it is a different regime, and consequential action lives only inside it.

No rule-binding by construction. An agent can be *told* a rule—a spending limit, an allowed counterparty set, a jurisdictional constraint—and may, probabilistically, respect it. There is no level on which the rule is binding rather than advisory. Guardrails written in the same medium as the behaviour they constrain are suggestions; what consequential delegation needs is a rule the agent *cannot* violate because the violation is rejected elsewhere, not merely discouraged within the model.

No proof of its own conduct. After the fact, an agent cannot demonstrate that it acted within its mandate; it can only assert it, and its assertion is exactly the kind of fluent, unfalsifiable artifact that machine generation now produces without cost. Detecting misconduct after the fact by inspecting outputs is, in the limit, unreliable: classifier-based detection of generated content decays toward chance as models improve [5], and strong watermarking admits generic removal under plausible assumptions [6]. Trust that must be reconstructed from the agent's own testimony is no trust at all.

No bounded, delegable authority. To hand an agent authority today is to hand it a credential—an API key, a wallet seed—that is effectively unbounded: whoever holds it can do whatever the underlying account can do. There is no native way to grant *this much and no more* such that the bound is enforced by the counterparty rather than hoped for by the principal. Delegation without an enforceable boundary is not delegation; it is abdication.

No settlement. An agent can compute that a transfer should happen; it cannot make it *have happened*. Finality—the point past which an action cannot be silently undone or reordered—is a property of a settlement system, not of a reasoner. Without it, every downstream party must price the risk that what the agent "did" will be revised, and that risk is a tax on every interaction.

No accountable identity. Asked "what is this, and who stands behind it?" an agent has no structural answer. It is not a person, holds nothing at risk, and its relationship to whatever principal deployed it is, to a counterparty, unverifiable. The oldest informal verification—ask, and watch the answers cohere—now selects for better generators, not honest actors.

These are not failures of cognition, and they do not shrink as cognition grows; a model twice as capable decides better and is no more bindable. They are failures of *consequence*: of the agent's inability to make its decisions rule-bound, provable, final, and attributable. Each is the mirror image of a property that one specific class of system has been engineering for a decade and a half.

3 The Deterministic Complement

The properties an agent lacks are the defining properties of a public ledger. This is not a coincidence to be admired but a complementarity to be used: the two systems are strong in exactly disjoint places. Generative models are open-ended, creative, and unconstrained, and they have no rules until someone supplies them; ledgers are rigid, rule-governed, and deterministic, and everything they do is about proof and provability. Neither is sufficient for autonomy that carries consequence. A ledger decides nothing—it is inert without an actor to propose actions. An agent settles nothing—it is unaccountable without a substrate to make actions bind. Put the generativity on one side and the bindability on the other (Figure 1a) and the design target is the upper-right region neither occupies alone: an agent that proposes, on a substrate that binds.

Read concretely, the substrate performs four functions the agent cannot. *It constrains before the fact.* An action can be made to carry a proof that it satisfies the principal's policy—this transfer is within budget, this counterparty is permitted, this disclosure is lawful—checked by the recipient before it takes effect, in the lineage of proof-carrying code, where a producer ships untrusted instructions together with a proof of their safety and the consumer checks the proof

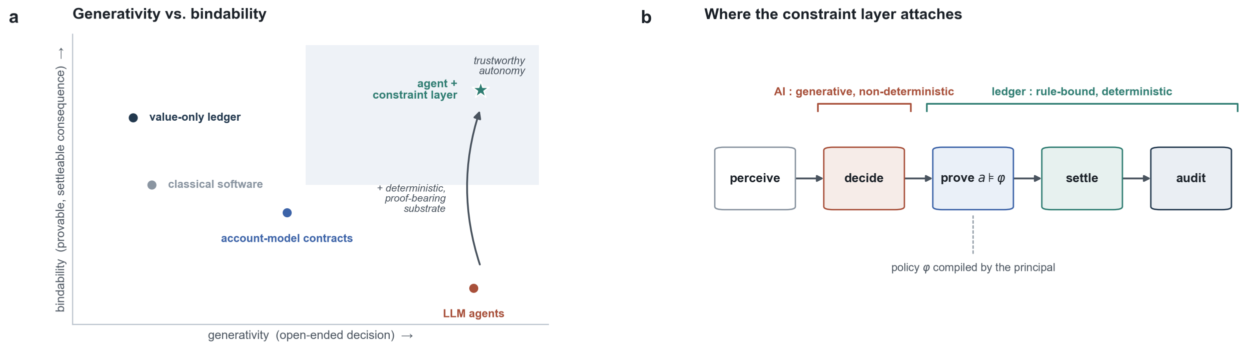

Figure 1: The complementarity. **(a)** Systems placed by *generativity* (capacity for open-ended decision) against *bindability* (capacity to make a decision provable, settleable, and final). Generative agents sit at high generativity and near-zero bindability; value-only and account-model ledgers sit at low generativity with partial bindability; the target—trustworthy autonomy—is reached only by pairing an agent with a deterministic, proof-bearing substrate. **(b)** The agent action lifecycle. The generative, non-deterministic step (*decide*) is the model's; the rule-bound, deterministic steps (*prove*, *settle*, *audit*) are the ledger's. The constraint layer attaches at the boundary, where a policy predicate φ compiled by the principal is enforced. Schematic.

Table 1: The division of labour. The agent's single native strength is the ledger's single native weakness, and conversely; consequential autonomy needs both columns at once.

Faculty	Autonomous agent	Public ledger
Open-ended decision	native	none
Determinism / replay	no	yes
Rule-binding by construction	advisory	rules are code
Proof of own conduct	assert only	proof-carrying
Irreversible settlement	no	yes
Bounded, delegable authority	unbounded	policy predicate
Accountable identity	none at stake	keys, public record

rather than trusting the producer [7]. The agent's authority becomes an object a stranger verifies, not a disposition the principal hopes for. *It settles.* On a deterministic substrate the action that was checked is the action that becomes final, with no gap for revision. *It turns compliance into a predicate.* Because the proof can be zero-knowledge, an agent can show it satisfies a rule without exposing the data behind it [8, 9]—over a jurisdictional threshold, holding sufficient collateral, an adult—so that settlement carries the evidence a counterparty needs and nothing more, and verification stops being surveillance. *It carries delegated authority natively.* The machinery for this is being standardised as we write: agent-payment protocols bind a purchase to a chain of cryptographically signed *mandates*—intent, then cart, then payment—so that authority rests on verifiable, non-repudiable evidence of what the principal sanctioned rather than on the model's probabilistic output [10]; an HTTP-native scheme lets an agent settle for a resource and attach the payment proof in a single exchange [11]; and account abstraction with session keys lets a principal mint a credential scoped to one counterparty, one amount, and a short validity, with the bound enforced at execution rather than trusted [12]. Smart contracts supplied the missing vocabulary years earlier—contractual clauses as executable objects [13]—and the drop in the cost of verification they enable is one of the primitive economies of the technology [14]. The point is not that any of this is new in isolation; it is that, assembled, it is exactly the faculty of consequence the agent is missing.

4 What a Ledger Must Be to Hold an Agent

"A ledger" is too coarse. A substrate meant to bind agents at scale must satisfy five requirements jointly; miss one and either the agent escapes the constraint or a trusted third party is quietly reintroduced one level down, with better typography. The requirements are stated from first principles; §5 reads a deployed system against them, and Table 2 states the mapping.

R1 — Provably secure consensus. A substrate whose job is to replace "trust me" with "check this" cannot rest its own security on "the operators seem honest." The base layer's safety must itself be a checkable claim, reduced to stated assumptions and an adversary model and proven, in the manner of the proof-of-stake protocols whose guarantees are established under explicit synchrony and corruption assumptions [15, 16]. A consequence layer secured by reputation is a contradiction in kind: it asks an agent's counterparties to do exactly what the layer exists to spare them.

R2 — Deterministic settlement: checked equals settled. The action an agent proves valid must be the action that settles. If a transaction's effect depends on global state at the moment of inclusion— on what else lands in the block and in what order—then the locally checked outcome and the globally settled outcome diverge, and the gap is harvested: front-running, sandwich attacks, value extracted between decision and finality [17]. For a human trader this is a cost; for an agent transacting at machine speed it is fatal, because the agent's proof of compliance was computed against a state that no longer holds at settlement. Local validation must equal global state transition (Figure 2b), or the constraint is checked against a world that does not arrive.

R3 — Expressive, privacy-preserving proofs. The substrate must express predicates, not merely transfers, and must let them be proven without disclosure—otherwise compliance is bought with surveillance, and an agent acting for a principal leaks the principal's position, counterparties, and strategy on every action. It must also compose: proofs about proofs, so that checking the system's own history does not itself become unbounded work. Foundations for private smart contracts make this concrete [20], and isomorphic off-chain execution lets the same rules run at scale and return as if computed on the base layer [19].

R4 — An economy without an owner. Someone must pay for ordering, storage, and availability. If a firm pays, the firm can censor and equivocate, and the constraint layer becomes a private database with extra steps—the trusted third party rebuilt. The substrate therefore needs an internal resource that buys security and operation from a permissionless population [22, 23]; this, not speculation, is the load-bearing role of a native asset. A verifier with an owner is a notary; a verifier with a self-funding commons is infrastructure.

R5 — Rules that can amend themselves. Predicates, proof systems, and parameters will change; cryptography ages and policy is contested. An immutable substrate ossifies; one amended by a foundation or a firm is owned (R4 again, one level up). The remaining option is constitutional: amendment through a recorded, rule-governed, on-chain process whose own legitimacy is itself checkable [24]. This is the least settled requirement anywhere, and the honest position is that it is an open institutional problem, not a solved engineering one.

Table 2: Five requirements for an agent-grade constraint layer: what each secures for a delegating principal, and the mechanism a deployed instance (§5) uses. References are to the primary design literature.

	Requirement	What it secures for the principal	Mechanism (case)
R1	provable consensus	the base itself need not be trusted	Ouroboros PoS [15, 16]
R2	deterministic settlement	the checked action is the settled one	extended-UTXO [18]
R3	private, composable proofs	compliance without leaking position	Hydra; Kachina/Midnight [19, 20]
R4	ownerless economy	no party can censor the agent	staking rewards [23]
R5	self-amending rules	the mandate language can evolve	on-chain constitution [24]

The requirements sort the field. A value-only ledger meets R1 and a strong form of R4 but fails R3 outright: it cannot express the predicates an agent's mandate is written in. The dominant account-model chains are expressive but surrender R2—their order-dependent execution is the native habitat of extractable value [17]—so an agent's checked action is not the action that settles. Permissioned and consortium ledgers recover determinism and expressiveness by abandoning R4: their validators are named and admitted, which is precisely the trusted third party an agent's counterparty would have to trust. And most designs are silent on R5. The five are easy to hold in pairs and hard to hold at once; that is why the existence of any joint solution is worth examining rather than assumed.

5 One Deployed Architecture

We read one system against R1–R5: Cardano, chosen because its documented design decisions—several of them long criticised as over-engineering—map onto the five with unusual completeness. The claim is architectural correspondence, not market success; the value of the case is that it converts "can the five be held jointly?" from a wish into an existence question with at least one positive instance. We take the requirements in turn and then state the case's liabilities plainly.

R1. The consensus layer is a family of proof-of-stake protocols with security proven under explicit adversary models—synchronous, then semi-synchronous with adaptive corruption [15, 16]. The methodological point outranks the cryptographic one: a layer that sells "check, do not trust" has made its own base-layer safety a checkable claim rather than an assurance, which is the discipline an agent's counterparty is relying on one level up.

R2. The accounting model extends Bitcoin's unspent-output discipline to full contracts while keeping its decisive property: a transaction's validity, effects, and fees are fixed by its own inputs, locally, before submission [18]. What the agent proves is what the chain settles. Account-based architectures, optimised for expressive shared state, give up exactly this, and the divergence between checked and settled is extracted industrially [17]. Under the present thesis determinism is not a performance taste; it is what makes an agent's action a provable claim about its own effect rather than a bid into an auction it cannot see.

R3. Expressiveness arrives in layers that do not tax non-users. On-chain scripts carry public predicates; state channels lift computation off-chain *isomorphically*, running the same scripts under the same semantics so a returned result is as if computed on the base layer [19]; and a dedicated partner network built on foundations for private smart contracts [20] runs predicates that prove compliance while disclosing only the predicate [21]. This is the selective-disclosure regime an agent mandate needs: prove the rule, withhold the position.

R4. Block production is bought from a permissionless population of stake pools under a reward scheme engineered to equilibrate at many independent operators [23], and protocol development is funded from an on-chain treasury. The substrate pays for its own neutrality with its own resource—the difference in kind between an open commons and the consortium attestation services now sold as enterprise blockchain, which have owners and therefore reintroduce the party an agent's counterparty must trust.

R5. Since 2023 the system has run an on-chain governance of delegated representatives, stake-pool operators, and a constitutional committee under a ratified constitution, with every motion on the public record [24]. The arrangement is young and its first constitutional crises have already arrived; that does not disqualify it, because R5 is unsolved everywhere, and what matters here is that the amendment process sits *inside* the verified record rather than in an adjacent boardroom.

An honest reading states the liabilities. Throughput on the base layer is well below centralised rails; the proof systems impose real prover and verification overhead; the application ecosystem has visibly struggled, and adoption lags the architecture by years; the governance of R5 is immature and contested. None of this is hidden by the thesis, and none of it is the thesis's subject: the case rests on joint satisfaction of the five *requirements*, and §7 states the evidence that would overturn the reading.

6 The Shape of Agent Commerce

The pieces compose into a single pattern (Figure 2a). A principal does not hand an agent a key; it compiles a policy φ—spending bounds, counterparty classes, escalation conditions—and the agent's authority *is* φ. Each action the agent takes carries a proof that the action satisfies φ; the counterparty verifies the proof before acting, so the boundary is enforced by the party protected by it, not by the principal's hope. Settlement occurs on deterministic rails, so the action verified is the action made final (R2); and because every step is recorded against the public ledger, audit is not a forensic reconstruction after a dispute but a replay of proofs already on the record. Authority is bounded *ex ante*; accountability is mechanical *ex post*.

This is not a forecast about how agents *might* transact; it is, in outline, how the first standards already say they will. The agent-payment protocols now shipping anchor each purchase to a chain of signed mandates precisely so that authorisation rests on verifiable intent rather than on a model's say-so [10, 11], and account abstraction supplies the bounded, revocable credential the mandate authorises against [12]. What they do not yet supply is the substrate guarantee of R2: most settle on account-model rails where the checked action and the settled action can still diverge [17]. The demand side of the complement is being built in the open; the determinism that would make it safe at machine speed is the half still missing, and §7 treats it as the live test it is.

Identity resolves along the same line. The question a counterparty must answer—*what is this, and who stands behind it?*—splits into checkable predicates. Personhood, where it is required, becomes something an actor can prove rather than assert [25]; agent credentials chain back to an accountable principal through signatures whose origin is itself checkable [26]; and the provenance of what an agent produces can travel with it as a signed claim rather than a guess for a detector to second-guess [27]. In each case the move is the same as the move that defines the whole architecture:

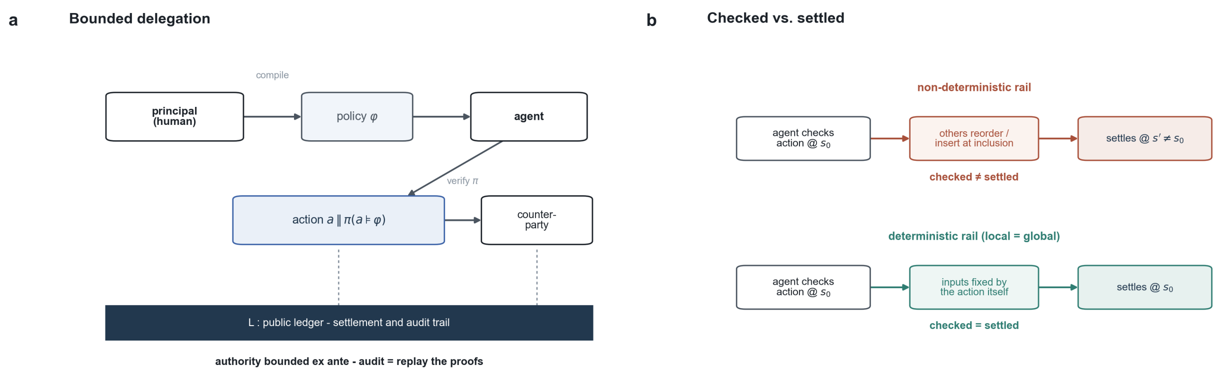

Figure 2: Bounded delegation and the determinism gap. **(a)** A principal compiles policy φ; the agent's action a travels with a proof $\pi(a \models \varphi)$ that the counterparty checks; settlement and audit reduce to replaying proofs against the public ledger L. Authority is bounded before the fact. **(b)** Why determinism (R2) is load-bearing: on a non-deterministic rail, reordering or insertion between check and inclusion settles a state $s' \neq s_0$, so the agent's verified action is not the one that settles; on a deterministic rail the action's inputs fix its effect, and checked equals settled. Schematic.

replace an after-the-fact judgement about a fluent artifact with a before-the-fact proof attached to it.

Two observations follow. The first is about cost. Humans experience this machinery as friction—keys, wallets, attestations, a worse experience than simply being trusted—which is why human-facing crypto has struggled to matter. Agents experience the opposite: reputation is a thing they cannot natively hold, while generating and checking proofs is what they do at native speed, and they do not tire, do not misread an address, and cannot be socially engineered through a substrate that admits only valid actions. The friction that repels human users is weightless to the population now arriving. The second is about direction of fit. Generative AI is the demand: it produces decisions faster than any trusted human apparatus can vet them. Verifiable, deterministic settlement is the only supply response whose throughput scales with compute rather than with attention. The two are not rivals competing for the same narrative slot; one makes decisions abundant, the other makes them bind—and a decision that cannot bind is, economically, just talk.

7 Open Problems and Disconfirmation

The thesis is a technical bet, and an honest bet names both its unsolved parts and the evidence that would settle it against us.

Four problems are open. *Cost and latency*: proving every action is not free, and an agent transacting continuously may generate proofs faster than they can be produced or verified cheaply; whether proof-bearing settlement can meet machine-speed commerce at acceptable cost is unresolved, and recursive composition (R3) is a partial answer, not a finished one. *Governance* (R5): self-amendment inside the verified record is young, thinly participated in, and already strained; no system has shown that a constitution can evolve a proof language and a fee policy without either capture or paralysis. *The predicate problem*: a mandate is only as good as the policy φ a principal can actually express, and writing φ for open-ended conduct— anticipating what an agent must never do—may be as hard as the alignment problem it resembles; the substrate enforces φ, it does not author it. *Bootstrapping*: the constraint layer is most useful once agents and counterparties already use it, the classic adoption trap, and architectural fitness does not dissolve it.

Four observations would disconfirm the account. The earliest agent-payment standards already settle on non-deterministic rails [11]; if agent commerce stays there at scale and over time, untroubled by the extractable-value tax [17], then R2 is wrong about what agents will pay to avoid, and determinism is not load-bearing. If provenance and agent identity consolidate into a handful of platform attestation services—device makers, app stores, model providers—then the ownerless substrate (R4) has lost to a recentralised one, and the testimony regime is simply rebuilt at higher concentration with root keys; this is, in our view, the likeliest way the thesis fails, and it is a question of market structure, not of feasibility. If advances make agents bindable *internally*—hardware-attested, verifiable execution carried within the model stack itself—then the external substrate is redundant and the complement dissolves. And if, among open substrates, the share of fees paid for settlement, proving, and attestation never overtakes the share paid for speculative turnover, then these systems are venues rather than infrastructure, whatever their design—a test the case of §5 is deliberately not exempt from.

8 Coda: Who Writes the Rule

The division of labour is the whole of the argument. A generative model supplies decision; a deterministic, proof-bearing, ownerless substrate supplies consequence; together they make an autonomous actor that can be delegated to, bounded, and held to account—which neither half can be alone. This is not a story about machines replacing institutions. It is a story about rebuilding, for non-human actors, the thing institutions did for human ones: convert a decision into a commitment that binds.

One faculty appears on neither side of the division. The substrate enforces the policy φ; the agent acts within it; but the choice of φ—what an agent must never do, what counts as compliant, which outcomes are worth making irreversible—is authored by neither. Writing the rule is not a task an agent can be delegated, because it is the act that defines the bounds of every delegation beneath it; and it is not a computation the substrate performs, because the substrate only checks. It is the authorship of constraint, and it stays with whoever bears the consequence of getting it wrong. As decision grows abundant and its settlement grows automatic, the scarce and consequential act is no longer deciding, and no longer enforcing. It is saying, in advance and on the record, what shall and shall not be allowed to bind.

Data availability. This is a theoretical contribution; it reports no experiment and no dataset. All claims rest on the cited literature.

Code availability. Both figures are schematic and are generated by the released script at `https://github.com/xinhao-zheng/constraint-layer`; no empirical series is plotted.

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Ethics note. No human subjects were involved. System and product names are used descriptively; the case analysis is architectural and is not investment guidance.

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